

Steven McDonald

Senior Frontend Engineer · Tech Lead

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Frontend engineer with 5 years leading the frontend of a clinical data platform serving 4,100+ daily users at VA hospitals and private medical groups as it grew from 3 to 15 customer sites. Previously managed international team of 3 engineers.

EXPERIENCE

HealthLevel — Mountain View, CA · Healthcare data management SaaS · 50 employees

Head of Data Visualization · Aug 2026 – Present, UI/UX Technical Lead · Oct 2023 – Present,
Technical Manager · Aug 2022 – Oct 2023, Software Engineer · Sep 2021 – Aug 2022

Architecture & Platform

- **Proposed and led migration of a 160K-line application** from a legacy Backbone/Mithril/jQuery stack to React and TypeScript. Designed a strangler pattern integration layer to render React components inside the legacy framework, allowing an incremental migration that avoided feature freeze and blocked no releases. Cut feature turnaround from ~2 months to ~2 weeks, and reduced new-engineer ramp from ~6 months to under 2 months.
- **Selected the state management and testing architecture** (Redux Toolkit, RTK Query, Zustand, Jest, React Testing Library) and migrated the codebase to strict TypeScript, authoring the typed contract between API responses and frontend consumers to eliminate recurring API integration failures.
- **Built the multi-tenant permissions layer** enforcing role-based access and data isolation across 15 customer sites, integrated with enterprise SSO/SAML and VA authentication.
- **Raised Lighthouse performance from 32 to 80** through route-level lazy loading, profiler-driven re-render elimination, and RTK Query cache tuning; cut CI build time from 15 to 10 minutes.

Product Work

- **Redesigned core flows to reduce clinician fatigue**, cut required clicks by 50%, added night mode, and reduced API calls by 30%.
- **Owned end-to-end delivery of a telestroke data portal for the National TeleStroke Program**, replacing spreadsheets and paper forms: ran physician interviews, iterated Figma mockups and prototypes, and shipped to production in 4 months.
- **Own visualization strategy for the platform**, and modernized a suite of Highcharts dashboards with KPI cards, bullet charts, and user-configurable views.
- **Architected a schema-driven forms engine powering customer onboarding** including conditional logic, multi-step wizards, autosave, conflict resolution, and inline validation — as well as a form builder and data-mapper tool built to support it.
- **Led WCAG 2.1 AA remediation of the full frontend** ahead of a federal Section 508 audit, a precondition for VA contracts, building accessible primitives verified with axe DevTools and VoiceOver.

Leadership

- **Led technical direction for 9 junior engineers and 1 UI designer**, interviewed 10 candidates, and previously managed 3 direct reports across US and India time zones with no attrition over two years.

On Me Gifting — San Francisco, CA · Gift card technology startup

Consultant · Aug 2023 – Oct 2023

- Built the first generation of the company's UI framework in Next.js and Tailwind CSS, and shipped cross-platform iOS and Android features in React Native.

PROJECTS

anycourseai.com · Apr – Aug 2026

- Schema-driven engine that generates and runs LLM-authored courses (16 published), with a RAG pipeline grounding content in college-level sources

the2senses.com · Mar – June 2021

- Next.js color blindness and hearing test suite for rural schools in India without access to medical facilities. Deployed by NGOs including HOPE Charitable Trust and used to screen 500+ students. Winner, "Tech for Good," FDM Technology Awards 2022.

EDUCATION

Cornell University — Ithaca, NY · B.S. Computer Science, cum laude Dec 2019

SKILLS

Languages TypeScript, JavaScript, Python, SQL, HTML5, CSS3, SASS

Frontend React, Next.js, React Native, Redux Toolkit, RTK Query, Zustand, React Hook Form, D3.js, Framer Motion, Tailwind CSS

Practice Jenkins, GitHub Actions, Git, Scrum, Design Systems, WCAG 2.1 AA, Figma, Jest, React Testing Library, Webpack, AI-assisted development (Claude Code, Copilot), HIPAA / Section 508 compliance, Data Visualization